

(1996) found that small changes in angina frequency were associated with minor changes in costs of illness but significant changes in WTP. Alberini and Krupnick (2000) estimated a WTP:COI ratio of 1.61 to 2.26 for symptoms associated with air pollution in Taiwan. Berger et al. (1987) interviewed a sample of 131 individuals in Denver and Chicago and found that for seven light health symptoms, mean daily consumer surplus estimates were always greater than mean daily COI estimates, by a factor of 3.1 to 79 times. A more recent study by Chestnut et al. (2006) compared COI and WTP values for preventing hospitalizations resulting from respiratory and cardiovascular illness. The study attempted to monetize a more comprehensive estimate of COI than is commonly used by including nontraditional cost components such as lost household production and lost recreation value due to the hospitalization and recovery period, in addition to medical expenses and lost earnings. They found that comprehensive COI estimates were much larger than previously obtained estimates that excluded nontraditional costs resulting from the hospitalization recovery period. Comprehensive COI estimates actually exceeded WTP values in nearly all cases.

In addition, a handful of studies have examined the relationship between WTP estimates derived from both the DBM and CVM, which can serve as a test of convergent validity. Dickie et al. (1986) collected data from a sample of 229 residents in the cities of Burbank and Glendora, California, to implement both methods, and compared WTP results across the two. The authors found that WTP estimates derived from the CVM were always larger than their DBM counterparts, by a factor of up to 10 times. However, a year later, Dickie et al. (1987) compiled a new dataset of residents in the same cities. In this survey, respondents were asked their WTP to avoid one day of recently experienced ozone-related symptoms. Each bid was then multiplied by the number of times symptoms occurred in a one month period and totaled across symptoms. Respondents then had a chance to revise their bid after seeing this total. Results from this study showed that average revised bids were much lower than original bids, and re-

vised WTP estimates from the CVM were found to be smaller than their DBM counterparts. Dickie et al. (1987) explained that this result is to be expected given that defensive goods used in calculations of WTP from the DBM may provide direct utility to the individual employing them. This should lead to larger benefit estimates than those derived from the CVM. Chestnut et al. (1996) found that CVM WTP estimates to avoid increases in angina were directly comparable to WTP estimates based on defensive expenditures.

As evident from the literature, there is still some uncertainty on the relationship between the estimates produced by the methods commonly used to value a reduction in health effects associated with exposure to a pollutant. Most importantly, while some studies have compared estimates across two of the three approaches used to value decreases in these morbidity effects, aside from Chestnut et al. (1996), none to our knowledge have attempted to compare estimates across all three methods using the same primary data. Such comparisons are needed to assess the validity of the techniques. Further, while the majority of studies compare point estimates of benefit measures, these comparisons can be made even more rigorous and accurate by evaluating statistical tests of their differences. Thus, we statistically test hypotheses about the equality of benefit estimates derived from the DBM, the CVM, and the COI approach.

Wildfire Smoke Exposure

The Centers for Disease Control and Prevention (CDC) and the EPA report that exposure to wildfire smoke can cause various ear, nose, and throat symptoms, as well as heightened symptoms in individuals with heart or lung disease. In addition, children and the elderly are considered sensitive populations whose health is at greater risk of being affected by wildfire smoke exposure. Evidence of these morbidity effects has been supported by studies such as those by Duclos, Sanderson, and Lipsett (1990), CDC (1999), Johnston et al. (2002), Mott et al. (2002), Kunzli et al. (2006), CDC (2008), and others, which all find a positive correlation between wildfire smoke exposure and various adverse

health effects and hospital admissions. Recent studies have called for the inclusion of their associated costs in damage assessments of a given wildfire (Butry et al. 2001; Morton et al. 2003; Abt, Huggett, and Holmes 2008; Dale 2009). However, the costs imposed on individuals as a result of these potential health effects are often unknown or underestimated.

While numerous studies have applied a COI approach to calculate the economic cost of health effects from wildfire smoke exposure specifically, to date, none have applied either the DBM or the CVM to calculate the WTP for a reduction in associated health damages. Kochi et al. (2010) conducted a literature review of studies estimating the economic cost of health damages from wildfire smoke exposure, and one conclusion was that understanding defensive actions taken to avoid exposure to the smoke should be studied, as their associated costs may be substantial.

IV. HYPOTHESIS

In this study, we compare the value of decreased morbidity from wildfire smoke exposure based on estimates from three different valuation methods: the DBM, the CVM, and the COI approach. As explained above, theory and empirical studies demonstrate that the traditional COI approach often underestimates the individual value for a reduction in health effects resulting from exposure to various pollutants. Bockstael and McConnell (2007) do derive a special case in which the COI estimate will equal the correct welfare measure for a change in pollution levels, but note that this is an extreme case. The conditions that need to hold for this equality are that increases in pollution do not cause any disutility, either directly or indirectly through illness, and that the individual cannot take defensive actions to mitigate the effects of changes in pollution concentrations. These conditions are unlikely to be true for wildfire smoke exposure. Past studies have consistently found that many residents in wildfire-prone areas know of the potential risks associated with wildfire smoke and take costly defensive actions to protect themselves against it (Mott et al. 2002; Kunzli et al. 2006; Kochi et al. 2010). Indeed, during

a major wildfire event, smoke advisory warnings are usually issued. These warnings provide specific suggestions for avoiding wildfire smoke. Incorporating nontraditional health costs such as the value of lost nonwork time in COI estimates may help to close the gap between COI estimates and WTP values, as noted by Berger et al. (1987) and Chestnut et al. (2006). In this study we will calculate a traditional COI estimate (COI_{trad}) as well as a comprehensive estimate that incorporates the value of lost recreation days resulting from an illness (COI_{comp}).

The expected relationship between WTP values for reduced morbidity estimated by the DBM and CVM remains unclear from both theoretical results and literature findings. For instance, as noted by Dickie et al. (1987), the defensive goods used in the DBM WTP calculation may provide a direct source of utility to the individual using them, meaning benefit estimates based on this method may be higher than their CVM counterparts. In addition, Carson et al. (1996) conducted a meta-analysis of revealed preference and CVM WTP estimates for quasi-public goods and found that on average, the former are larger than the latter. However, under the DBM, equation [8] shows that the WTP for a marginal reduction in symptom days is equal to the savings in defensive expenditures that just offsets the change in sick days, holding environmental quality constant. However, it is unclear that using actual observed defensive expenditures to estimate this value will hold environmental quality constant, meaning WTP estimated by the DBM could underestimate true WTP (see Courant and Porter [1981], Harrington and Portney [1987], and Bockstael and McConnell [2007], for a derivation of this argument in the context of WTP for changes in environmental quality). Therefore, the hypothesis we test is as follows:

$$H_0: COI_{trad} = COI_{comp} = WTP_{DBM} = WTP_{CVM},$$

$$H_a: COI_{trad} < COI_{comp} < WTP_{DBM} \neq WTP_{CVM}.$$

While the COI estimates are simply a sample mean, the WTP values will be calculated through more complex mathematical formulas and will not have straightforward proper-

ties that facilitate simple statistical comparison of the measures. Therefore, two approaches are implemented to test this hypothesis. First, the bootstrapping method (see Efron 1979, 1982; Efron and Tibshirani 1993) will be applied to draw a new sample with replacement from each original dataset used to estimate WTP values. This process will be repeated 1,000 times to generate a distribution of 1,000 values calculated from each of the two WTP methods.¹ Percentile confidence intervals will then be constructed. To do so, the distribution of values is first ordered from the lowest to the highest value and then to form, for example, for the 90% confidence interval, 5% of the observations at each tail are dropped from the distribution. As explained by Loomis, Creel, and Park (1991), while benefit estimates obtained using different valuation methods may appear to be quite different, comparing confidence intervals is a first step to statistically test for a difference.

If the confidence intervals around two of the values do not overlap, it can be concluded that the null hypothesis for that comparison can be rejected at the specified level of confidence. If the confidence intervals of any two values do overlap at the desired level of confidence, a second approach will be applied. Following Poe, Severance-Lossin, and Welsh (1994) and Poe, Giraud, and Loomis (2005), we use a complete combinatorial approach based on the method of convolutions. This unbiased, nonparametric test is used to evaluate the statistical difference between two simulated distributions by generating a third distribution consisting of all possible differences between the two distributions of interest. For instance, if comparing WTP_{DBM} with WTP_{CVM} , this third distribution is constructed by calculating $((WTP_{DBM})_i - (WTP_{CVM})_z)$, where $i = 1,000$ bootstrapped WTP_{DBM} values and $z = 1,000$ bootstrapped WTP_{CVM} values. This results in a 1,000,000 by 1 vector of dif-

ferences. The proportion of negative values from this third distribution of differences represents the probability for the two WTP distributions to be overlapping (if this value is greater than 0.5, it should be subtracted from 1). This probability represents the one-sided p -value associated with the hypothesis test of equality for the two distributions. Multiplying this value by two gives the two-sided p -value associated with this test. If this p -value is less than a particular level of significance, say, $\alpha = 0.05$, the null hypothesis of equality between the two WTP estimates can be rejected at the 0.05 significance level.

V. SAMPLING FRAME AND DATA

The Station Fire began on Wednesday, August 26, 2009, in the Angeles National Forest, located adjacent to the Los Angeles, California, metropolitan area. It took firefighters 52 days to fully contain the wildfire, which crept into nearby neighborhoods. This became the largest wildfire in Los Angeles County's modern wildfire history and the tenth largest in California's. The smoke from the Station Fire caused nearby residents to experience unhealthy air quality levels, and as a result, smoke advisory warnings were issued by the South Coast Air Quality Management District. These warnings advised residents in any area impacted by smoke to avoid unnecessary outdoor activities, keep windows and doors closed when indoors, and run the air conditioner.

The sample frame for the study was 1,000 randomly selected residents living in five southern California cities (Duarte, Monrovia, Sierra Madre, Burbank, and Glendora) that were exposed to unhealthy air quality levels due to smoke from the Station Fire. A questionnaire was created in the summer of 2009 based on feedback from focus groups held in Anaheim, California, and consultation with health economists. The focus groups allowed participants to pretest and critique the questionnaire based on their experience with the Freeway Complex Fire of 2008.

Approximately six weeks after the Station Fire began, the survey was mailed to the sample of southern California residents. At that time, the wildfire was 99% contained and re-

¹ While other simulation methods including the jackknife, Krinsky and Robb (1986), Cameron (1991), and delta methods could also be applied, studies comparing across methods have found that they are all relatively accurate and will produce similar results. See Cooper (1994) and Hole (2007) for these findings and an explanation of when the methods will differ.

spondents who had evacuated would have had a chance to return home. Two follow-up mailings were implemented over a two-month period, for a total of three mailings to nonrespondents. Forty surveys were not deliverable, and 458 complete surveys were returned for an overall response rate of 48%. After removing incomplete surveys and surveys from respondents who were not home during the fire, there remain a total of 413 usable surveys.

Data on concentrations of particulate matter (PM) and carbon monoxide (CO) released during the Station Fire were taken from the California Environmental Protection Agency Air Resources Board. Of the five cities surveyed for the study, Burbank and Glendora are the only two with air quality monitoring stations within city limits. However, the other cities have stations close by. Data on PM_{2.5} concentrations during the weeks the wildfire burned were available for the cities of Burbank and Glendora, while data on PM₁₀ concentrations were available for the city of Glendora only. Data on CO were directly available from monitoring stations in Burbank and Glendora. While there are no monitoring stations in Duarte, Monrovia or Sierra Madre, there are monitoring stations very close by, which reported levels of CO during the weeks the Station Fire burned. CO concentrations from the Azusa monitoring station were used as a proxy for levels in Duarte and Monrovia, as the station is located four miles from the former and six miles from the latter. CO concentrations from the Pasadena monitoring station were used as a proxy for levels in Sierra Madre, as these cities are located six and a half miles apart. Figure 1 provides an approximate location of the cities surveyed and air quality monitoring stations where data were used, in relation to the location of the Station Fire.

Given recent findings by Kunzli et al. (2006) that subjective, within-community pollution measures may reflect wildfire smoke exposure more accurately than objective community-wide measures, the survey questioned respondents about the smoke and/or ash they smelled both inside and outside of their home for the weeks following the start of the fire. If they indicated that they could smell smoke and/or ash, they were asked about the number

of days that it was smelled. Table 1 presents both objective and subjective pollution levels by city during the Station Fire. Objective pollution levels are measured by six-day averages of daily maximum and daily average concentrations of PM_{2.5}, PM₁₀, and CO, where data were available. The day the Station Fire began and the five days following, the period from August 26, 2009, through August 31, 2009, represents the peak period, as concentrations tapered off considerably after this six-day window. Table 1 also presents the number of survey respondents in each surveyed city, and within each city the number and percentage of respondents who smelled smoke and/or ash for the number of days specified in each given range.

To implement the DBM, additional data were gathered on the length of time health symptoms were experienced as a direct result of exposure to the wildfire smoke, as well as the type of symptom experienced. Respondents were asked to report the number of days symptoms were experienced by themselves as well as all household members. Respondents were then presented with a table of potential defensive actions that could have been taken to reduce their exposure to the wildfire smoke. The list of potential defensive actions were chosen based on a number of factors, including recommendations from the CDC on what to do during a fire to decrease exposure to the smoke, recommendations by focus group participants, and finally, the actions that previous studies have confirmed individuals take during wildfires to decrease their exposure (Mott et al. 2002; Kunzli et al. 2006). Given that many defensive actions, such as running an air conditioner, can inherently be taken for reasons other than to prevent wildfire smoke exposure, respondents were specifically told to indicate whether they had taken each action as a direct result of exposure to the wildfire smoke and for no other reason. Respondents were also asked to report the cost of taking each action where this information was applicable. Table 2 outlines the number and percentage of survey respondents who reported taking each defensive action, along with the average cost reported by those respondents who took that action. Eighty-nine percent of respondents took at least one defensive action.